

Problem

A 240-V rms 60-Hz source supplies a parallel combination of a 5-kW heater and a 30-kVA induction motor whose power factor is 0.82. Determine:

(a) the system apparent power

(b) the system reactive power

(c) the kVA rating of a capacitor required to adjust the system power factor to 0.9 lagging

(d) the value of the capacitor required

Step-by-step solution

Step 1 of 10

(a)

The real power of heater is,

$$P_1 = 5 \text{ kW}$$

The reactive power of heater is zero.

$$Q_1 = 0 \text{ VAR}$$

Calculate the complex power of heater.

$$\begin{aligned} S_1 &= P_1 + jQ_1 \\ &= 5 \text{ kVA} \end{aligned}$$

Step 2 of 10

Calculate the value of phase angle of inductor motor.

$$\cos \theta_2 = \text{pf}_2$$

$$\cos \theta_2 = 0.82$$

$$\theta_2 = \cos^{-1}(0.82)$$

$$\theta = 34.9^\circ$$

Calculate the real power of inductor motor.

$$\begin{aligned} P_2 &= S_2 \cos \theta_2 \\ &= (30 \times 10^3)(0.82) \\ &= 24.6 \text{ kW} \end{aligned}$$

Step 3 of 10

Calculate the reactive power of inductor motor.

$$\begin{aligned}Q_2 &= S_2 \sin \theta \\&= (30 \times 10^3) \sin(34.9^\circ) \\&= (30 \times 10^3)(0.572) \\&= 17.16 \text{ kVAR}\end{aligned}$$

Calculate the complex power of inductor motor.

$$\begin{aligned}S_2 &= P_2 + jQ_2 \\&= 24.6 + j17.16 \text{ kVA}\end{aligned}$$

Step 4 of 10

Calculate the total complex power of the system.

$$\begin{aligned}S &= S_1 + S_2 \\&= (5 \text{ kVA}) + (24.6 + j17.16 \text{ kVA}) \\&= 29.6 + j17.16 \text{ kVA}\end{aligned}$$

The real power of the system is,

$$P = 29.6 \text{ kW}$$

The reactive power of the system is,

$$Q = 17.16 \text{ kVAR}$$

Step 5 of 10

Calculate the value of apparent power of the system.

$$\begin{aligned}S &= \sqrt{P^2 + Q^2} \\&= \sqrt{(29.6 \times 10^3)^2 + (17.16 \times 10^3)^2} \\&= 34.2 \times 10^3 \\&= 34.2 \text{ kVA}\end{aligned}$$

Thus, the value of apparent power of the system is 34.2 kVA.

Step 6 of 10

(b)

The total complex power of the system is,

$$S = 29.6 + j17.16 \text{ kVA}$$

The reactive power of the system is,

$$Q = 17.16 \text{ kVAR}$$

Thus, the value of reactive power of the system is 17.16 kVAR.

Step 7 of 10

(c)

Calculate the power factor of the system.

$$\begin{aligned}\text{pf} &= \frac{P}{S} \\ &= \frac{29.6 \times 10^3}{34.2 \times 10^3} \\ &= 0.865\end{aligned}$$

Calculate the value of phase angle of the system.

$$\begin{aligned}\cos \theta &= \text{pf} \\ \cos \theta &= 0.865 \\ \theta &= \cos^{-1}(0.865) \\ \theta &= 30^\circ\end{aligned}$$

Step 8 of 10

Calculate the value of new phase angle of the system.

$$\begin{aligned}\cos \theta_{\text{new}} &= \text{pf}_{\text{new}} \\ \cos \theta_{\text{new}} &= 0.9 \\ \theta_{\text{new}} &= \cos^{-1}(0.9) \\ \theta_{\text{new}} &= 25.84^\circ\end{aligned}$$

The real power for the new power factor should not change.

$$\begin{aligned}P_{\text{new}} &= P \\ &= 29.6 \text{ kW}\end{aligned}$$

Step 9 of 10

Calculate the reactive power of new system.

$$\begin{aligned}Q_{\text{new}} &= P_{\text{new}} \tan \theta_{\text{new}} \\ &= (29.6 \times 10^3) \tan(25.84^\circ) \\ &= (29.6 \times 10^3)(0.484) \\ &= 14.33 \text{ kVAR}\end{aligned}$$

Calculate the kVA rating of the capacitor.

$$\begin{aligned}Q_C &= Q - Q_{\text{new}} \\ &= 17.16 \text{ kVAR} - 14.33 \text{ kVAR} \\ &= 2.83 \text{ kVAR}\end{aligned}$$

Thus, the kVA rating of the capacitor is **2.83 kVAR**.

Step 10 of 10

(d)

Calculate the value of the required capacitor to adjust the system power factor.

$$\begin{aligned}C &= \frac{Q_C}{\omega V_{\text{rms}}^2} \\&= \frac{Q_C}{2\pi f V_{\text{rms}}^2} \\&= \frac{2.83 \times 10^3}{2\pi (60)(240)^2} \\&= 130.32 \times 10^{-6} \\&= 130.32 \mu\text{F}\end{aligned}$$

Thus, the value of the required capacitor to adjust the system power factor is 130.32 μF .